

Grayscale Image Quantization Report

Quantization is the process of mapping continuous intensity values into discrete levels. It reduces image detail when fewer levels are used and introduces banding effects.

Input Image (256 Levels):



Python Code:

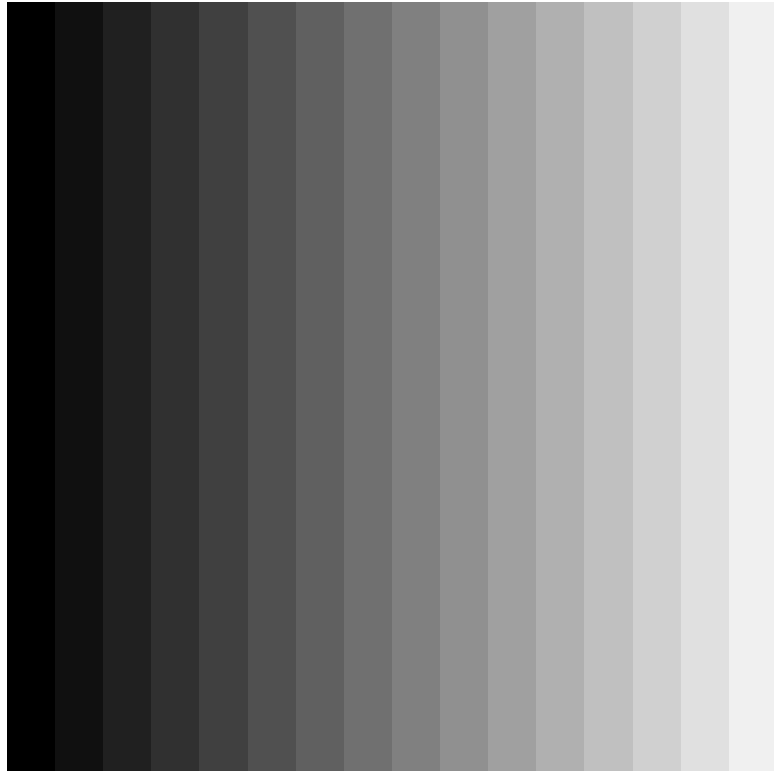
```
# Load image
img = cv2.imread('image.jpg', cv2.IMREAD_GRAYSCALE)

# Quantization function
def quantize(image, levels):
    step = 256 // levels
    return (image // step) * step

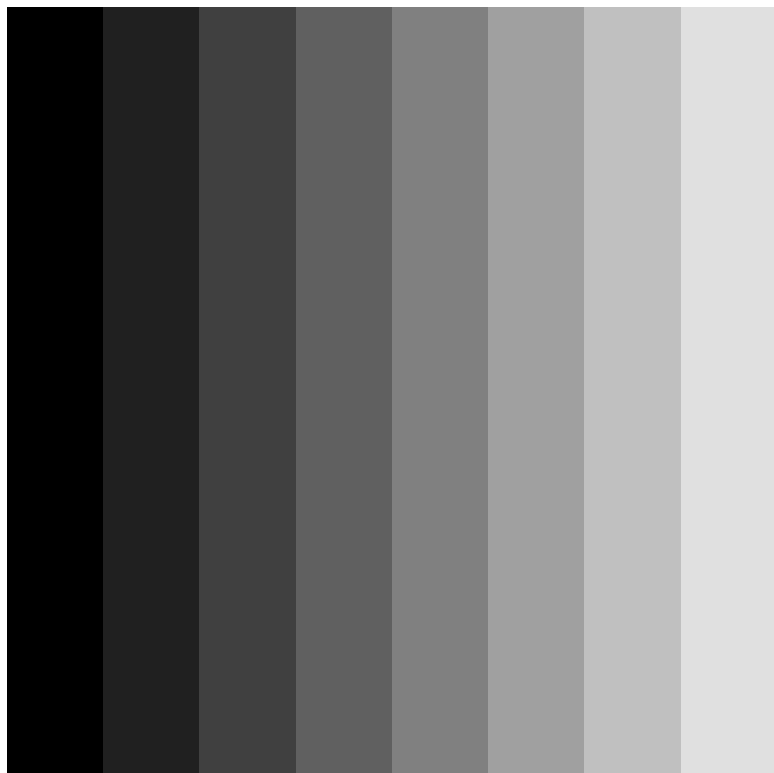
# Apply quantization
q16 = quantize(img, 16)
q8 = quantize(img, 8)
q4 = quantize(img, 4)
```

Output Images:

16 Levels:



8 Levels:



4 Levels:

